Smart LPG Gas Safety & Leak Alert System

Developed by Anant Jaiswal / AJ_ORBIT

1. Pin Connections (Arduino Nano)

MQ-5 Gas Sensor (Detects Gas Leak):
- VCC -> 5V on Nano
- GND -> GND on Nano
- A0 -> A0 on Nano (Analog Input)

High-Torque Servo Motor (Turns off Regulator):
- VCC (Red) -> Positive terminal of External Battery
- GND (Brown/Black) -> GND on Nano AND Negative terminal of Battery
- Signal (Yellow/Orange) -> D9 on Nano

SIM800L GSM Module (Makes the Phone Call):
- VCC -> 4V Power Supply (Buck Converter)
- GND -> GND on Nano
- TXD -> D2 on Nano (Nano RX)
- RXD -> D3 on Nano (Nano TX) ***SEE WARNING BELOW***

Relay Module & Buzzer (Sound Alarm):
- VCC -> 5V on Nano
- GND -> GND on Nano
- IN (Signal) -> D4 on Nano
- Relay Output -> Connect Buzzer in series with battery through COM and NO.

Regulator Switch (Starts the 10-min Timer):
- Terminal 1 -> D5 on Nano
- Terminal 2 -> GND on Nano

2. IMPORTANT HARDWARE WARNINGS

VOLTAGE DIVIDER REQUIRED FOR SIM800L:
The Arduino Nano sends out 5V on its TX pin, but the SIM800L's RX pin expects 3.3V.
Connecting them directly will burn out the SIM800L module over time.

The Fix (Voltage Divider):
1. Place a 1k Ohm resistor between Nano D3 (TX) and SIM800L RX.
2. Place a 2k Ohm resistor from the SIM800L RX to GND.

Note: The SIM800L TX pin sends ~3.3V to the Nano RX (D2). The Nano reads 3.3V

perfectly fine, so that specific line can be connected directly without resistors.

POWER WARNING FOR SERVO & GSM:

Do NOT power the high-torque servo or the SIM800L directly from the Arduino Nano's 5V pin. They draw high peak currents (up to 2 Amps) that will fry the Arduino's voltage regulator. Always use an external battery or buck converter to power them.